

YouTube Niche Research

A simple guide to setting it up and what each tool is for

What is this?

A free desktop app that helps a YouTube creator decide **what to make next**. You run it on your own computer. It reads public YouTube data and uses AI to turn that into plain-English advice — what's working in a niche, what to copy from a specific video, and which topics are trending right now. Its whole design goal is to be **honest**: every number comes with the context that makes it mean something, so you're never misled by a stat that just looks impressive.

Who it's for

Creators and researchers who want real signal, not vanity numbers. You don't need to be a programmer — if you can copy files into a folder and run two commands, you can use it.

The three tools

Niche / Genre Analysis

What it does: Pulls recent videos for a topic, separates Shorts from long-form, and shows what's over- and under-performing plus what the top videos do differently. Saves each run so it can tell real patterns from random noise over time.

Use it to: Understand a niche and find the patterns that actually hold up.

Single-Video Teardown

What it does: Analyses one video: how it did versus the creator's own channel, an AI 'what to steal', a plain summary of what it's about, the moments the audience flagged in the comments, and which parts could become Shorts.

Use it to: Learn exactly what to copy from a video — and what to clip.

Trend Discovery

What it does: Pick a broad area (Gaming, Cooking, Fitness...) and it finds the specific trending sub-niches inside it, each with its most popular videos and a label showing whether it's a real multi-channel trend or just one viral video.

Use it to: Spot rising, under-served topics worth jumping on.

Set it up in 4 steps

You'll need **Python 3** installed. Everything below is typed into a terminal in the project folder.

1. Install the dependencies

```
pip install -r requirements.txt
```

2. Add your API keys. Make a file named `.env` in the same folder as `app.py`, with two lines:

```
YT_KEY=your_youtube_data_api_key  
  
ANTHROPIC_API_KEY=your_anthropic_key
```

3. (Optional) Check everything's healthy — no keys needed:

```
python self_check.py
```

4. Launch it — opens in your browser at localhost:8501:

```
streamlit run app.py
```

Where to get the two keys

YT_KEY	YouTube Data API v3 key from the Google Cloud Console. The free tier (10,000 units/day) is plenty — a typical analysis costs a few hundred.
ANTHROPIC_API_KEY	From the Anthropic Console (console.anthropic.com). Powers the AI summaries, teardowns and clip ideas.

The app opens and lets you click around without keys — it just can't fetch live data until they're set. Keep your keys private; never put them in a public repo.

Why you can trust the numbers

Age-fair comparisons. A new video and an old one can't be judged on raw views — the old one had more time. This app always compares a video to others of a **similar age**, so 'what's working' isn't just 'what's newest'.

It admits what it can't know. Clip ideas come from the transcript's content, not secret 'most-replayed' data (which YouTube doesn't share). 'Risk' notes are hints to check, not a copyright verdict. A trend that's really one channel is labelled as exactly that.

Tip: if a transcript won't auto-load, YouTube is blocking it — just paste the transcript into the box (the YouTube (...) menu -> Show transcript → copy). Everything else still works.